Subject: Mathematics **CBSE CLASS 10 – CHAPTER TEST (SET – 3)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 7: Coordinate Geometry** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The distance of point $P(3a, 4a)$ from y-axis is : [1]

(A) $3a$ (B) $-3a$ (C) $4a$ (D) $-4a$

Q2. The distance of point $(a, -b)$ from x-axis is : [1]

(A) a (B) $-a$ (C) b (D) $-b$

Q3. The distance of the point $A(-3, -4)$ from x-axis is : [1]

(A) 3 (B) 4 (C) 5 (D) 7

Q4. The point on y-axis equidistant from the points $A(1, 3)$ and $B(4, 4)$ is : [1]

(A) $(0, 11)$ (B) $(11, 0)$ (C) $(0, 13)$ (D) $(0, 12)$

Q5. The perimeter of a triangle with vertices $(0, 4)$, $(0, 0)$ and $(3, 0)$ is: [1]

(A) 5 units (B) 12 units (C) 11 units (D) $7 + \sqrt{5}$ units

SECTION B: Short Answer Type Questions (6 Marks)

Q6. The distance of a point A from x-axis is 3 units. Which of the following cannot be coordinates of the point A ? (a) $(1, 3)$ (b) $(-3, -3)$ (c) $(-3, 3)$ (d) $(3, 1)$ [3]

Q7. Find a relation between x and y such that $P(x, y)$ is equidistant from the points $A(3, 5)$ and $B(7, 1)$. Hence, write the coordinates of the points on x-axis and y-axis which are equidistant from points A and B. [3]

SECTION C: Long Answer & Competency Based Questions (9 Marks)

Q8. $P(x, y)$, $Q(-2, -3)$ and $R(2, 3)$ are the vertices of a right triangle PQR right angled at P. Find the relationship between x and y. Hence, find all possible values of x for which $y = 2$. **[5]**

Q9. Competency Case Study: In a classroom, 4 friends are seated at the points A, B, C and D as shown in the Cartesian grid. Champa and Chameli walk into the class and after observing for a few minutes Champa asks Chameli, "Don't you think $ABCD$ is a square?" Chameli disagrees. The coordinates are $A(3, 4), B(6, 7), C(9, 4)$ and $D(6, 1)$. **[4]**

(i) What is the distance AB?

(ii) What is the distance BC?

(iii) What is the length of the diagonal AC?

(iv) Is $ABCD$ a square or a rhombus but not a square?

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